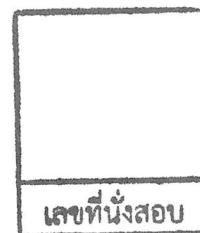


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 2

Year: 2017

Subject: 392152 Chemistry II

Section: 15-18

Date: 18 May 2018

Time: 10.00-12.00

Name: _____ ID: _____ Section: _____

Instructions:

1. The examination has 8 pages (including this page), 2 sections and a total score of 48 points.
 2. Write all your solution and answer on this examination sheet.
 3. This is a closed book examination.
 4. You are not allowed to leave the exam room during the first hour after the beginning of the exam.
 5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
 6. You are not allowed to use restroom during the exam except in case of an emergency.
 7. No documents are allowed to be taken out of the examination room.
 8. Calculators are **NOT** allowed in the examination.
 9. Electronic communication devices are **NOT** allowed in the examination room.
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**Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University.**

Solvent	Boiling Point (°C)	K _b (°C/m)	Freezing Point (°C)	K _f (°C/m)
Water	100	0.512	0	1.86
Benzene	80.1	2.53	5.5	5.12

	Multiple Choice	Part 2.1	Part 2.2	Part 2.3	Part 2.4	Total
score						

Part 1. Multiple Choice (16 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				

Question no.	a.	b.	c.	d.
9				
10				
11				
12				
13				
14				
15				
16				

1. Which one is incorrect?
 - a. Stock solution is a high concentration solution
 - b. Dilution is a process that use for decreasing the concentration of solution
 - c. Volumetric flask is used to measure the volume of the solution
 - d. None of it
2. Which one is correct?
 - a. Colligative properties are the chemical properties of solution
 - b. Colligative properties depend on type of solute
 - c. Colligative properties depend on the concentration of solute in the solution
 - d. b. and c. are correct
3. Which one is incorrect about osmotic pressure?
 - a. Osmotic pressure is colligative properties
 - b. Semipermeable membrane allows solute to pass through it
 - c. Osmotic pressure increases with increasing temperature
 - d. Osmotic pressure decreases with decreasing of M
4. What volume of 15 M sulfuric acid must be used to prepare 1 L of a 1.5 M H_2SO_4 solution?
 - a. 0.1 mL
 - b. 1.0 mL
 - c. 10 mL
 - d. 100 mL
5. According to an empirical formula's definition, which choice is incorrect?
 - a. CO_2
 - b. HO
 - c. H_2
 - d. H_2O
6. Which statement is true?
 - a. Formula which gives simplest whole number ratio of atoms is molecular formula.
 - b. Empirical formula and molecular formula may be the same in certain cases.
 - c. Molecular formula are always different from empirical formula.
 - d. all of the above
7. Which statement is wrong about hydrogen peroxide (H_2O_2)?
 - a. Molecular mass is equal to 34 g/mol.
 - b. It is composed of 94.12% oxygen.
 - c. Ratio between oxygen and hydrogen atoms is 1:1.
 - d. Empirical formula is H_2O_2 .

8. The empirical formula for a hydrocarbon is found to be CH_2 . In a separate experiment, the molar mass is determined to be approximately 112 g mol^{-1} . The number of hydrogens in the molecular formula of the hydrocarbon is therefore
- a. 8 b. 12 c. 16 d. none of these
9. Zinc metal reacts with hydrochloric acid by the following reaction: $\text{Zn} + \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
- What is the limiting reactant, if 2 mol of Zn is added to HCl containing 5 mol?
- a. H_2 b. HCl c. Zn d. ZnCl_2
10. In problem 9, how many grams of the excess reactant remain after the reaction is complete?
- a. 18.3 g b. 36.5 g c. 65.5 g d. 73.0 g
11. In problem 9, how many moles of ZnCl_2 are produced?
- a. 0.5 mol b. 1.0 mol c. 1.5 mol d. 2.0 mol
12. The actual yield of a reaction is 33 g and the theoretical yield is 50 g. What is the percentage yield?
- a. 66 % b. 78 % c. 87 % d. 90 %
13. The s-block elements are _____.
- a. transition elements b. main group elements
c. inner transition elements d. alkaline metal elements
14. Oxides of most nonmetals combine with water to form _____.
- a. an acid b. a base c. water and a salt d. water
15. A halogen atom _____.
- a. has one electron in its outer shell b. has two electron in its outer shell
c. has a felled outer shell d. lacks one electron of having a felled out shell
16. The place of an element in the periodic table is determined by its _____.
- a. atomic number b. atomic mass c. density d. chemical activity

Part 2 Write your answer in the space provided for each question (32 points)

Dilution of solution and colligative properties

1.1 How many mL of 4.0 M NaCl is needed to make 50 mL of a 0.5 M NaCl solution? (2 pts)

[illegible]

1.2 How much water would need to be added to 500 mL of a 2.5 M HCl solution to make a 1.0 M solution? (2 pts)

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1.3 3.0 mol of NaCl are added to 500 g of H₂O. What is the freezing point and boiling point of this solution? (4 pts)

[illegible]

Empirical/Molecular Formula

2.1 30 g of a compound containing only C and H gave on complete combustion 88 g of CO_2 and 54 g of H_2O . Calculate the empirical formula of the compound. (4 pts)

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2.2 A compound was found to contain 48 g carbon and 10 g hydrogen. The molecular mass of the compound is 58 g/mol. Determine the molecular formula. (4 pts)

[illegible]

Limiting reagent and Yield percentage

3.1 In an experiment (see the reaction below), 6.80 g of NH_3 are reacted with 6.40 g of O_2 .



- Which reactant is the limiting reagent? (2 pts)

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- How many grams of NO are formed? (1 pt)

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- How many grams does the excess reactant remain after the reaction? (1 pt)

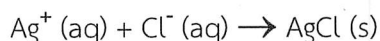
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3.2 A solution containing silver ion, Ag^+ , has been treated with excess of chloride ions Cl^- :



When dried, 0.1200 g of AgCl has recovered. Assume the percentage yield to be 75%,

Calculate **theoretical yield of AgCl**? (1 pt)

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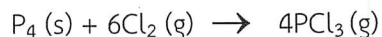
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3.3 When Phosphorus reacts with Chlorine it forms Phosphorus Trichloride:



- How much PCl_3 can be formed by the reacting 6.0 g of P_4 with excess Chlorine? (2 pts)

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 - If a 19.17 g sample of PCl_3 was obtained from the reaction, calculate the percent yield. (1 pt)

Review of periodic table and oxide/chloride compounds

4. There are 4 elements which are A, B, C, and D. Their atomic numbers are 4, 6, 14 and 19, respectively. Write electron configurations for their elements and complete the following table. (8 pts)

A =,

B =

C =,

D =

	A		B		C		D	
	Oxide	Chloride	Oxide	Chloride	Oxide	Chloride	Oxide	Chloride
Chemical formula								
Acid-base behavior								

Jarukorn Sripradite

Parinya Kongprom

Sunanta Chuayprakong

Monrudee Phongaksorn